Study-aware QSPR of human skin permeability: separating molecular signal from source heterogeneity and calibrating unseen sources

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1 Abstract

Human skin permeability measurements are commonly pooled across studies even though permeability depends not only on molecular properties but also on study-specific experimental conditions. We tested whether this heterogeneity can be modeled as a reusable source-associated component rather than treated solely as noise. Three public human-skin permeability resources were used sequentially rather than pooled: the Cheruvu et al. 2022 dataset for development, HuSkinDB for replication in a separately curated database and held-out-source calibration, and SkinPiX for cross-dataset transfer. Measurement-level mixed-effects models separated molecular, residual compound-specific, literature-source, and residual variation. The molecular component was deliberately simple and prespecified, using molecular weight and logP alone (S1) or a six-descriptor physicochemical specification (S2). In Cheruvu et al., source-associated variation accounted for 22.8% (95% CI 14.4–31.9%) of modeled random/residual variance; in HuSkinDB it accounted for 44.4% (33.5–54.5%). Source-blind hierarchical prediction did not show a reproducible improvement over conventional aggregation: RMSE improvement was 5.4% (95% CI -3.6–10.9%) for Cheruvu S1, 7.0% (-5.1–14.0%) for Cheruvu S2, and -1.7% (-5.1–1.0%) in HuSkinDB. In contrast, source-aware hierarchical prediction improved RMSE by 16.7% (11.6–21.8%), 18.7% (11.5–23.3%), and 17.0% (10.0–23.8%), respectively. When entire HuSkin sources were held out, offsets estimated from shared anchor compounds improved prediction of compounds absent from model fitting; among 9 matched sources, mean source-level RMSE reduction increased from 31.1% (13.4–42.1%) with 1 anchor to 45.3% (19.2–60.0%) with 5 anchors. A pairwise double-holdout falsification showed that source-matched calibration consistently outperformed an equivalently estimated offset from another source. Finally, one-anchor calibration reduced mean source-level RMSE by 17.6% across 4 eligible SkinPiX sources containing 54 HuSkin-unseen compounds, all within the HuSkin-defined chemical applicability domain. These findings support a study-aware QSPR framework in which literature-source heterogeneity is modeled explicitly and can be estimated for a new experimental source using a small number of shared compounds.

Keywords: skin permeability; QSPR; mixed-effects model; hierarchical model; source calibration; external transfer; applicability domain; HuSkinDB; SkinPiX

2 Introduction

The skin permeability coefficient, K_p, is a central parameter in transdermal drug delivery, dermal toxicology, cosmetics, and exposure assessment. Because direct measurement in human skin is expensive, tissue-limited, and experimentally demanding, quantitative structure–property relationship (QSPR) models have long been used to estimate permeability from molecular structure. The classical Potts–Guy model relates permeability primarily to lipophilicity and molecular size (Potts and Guy 1992), while later studies have explored broader

physicochemical descriptor sets and more flexible statistical or machine-learning methods (Tsakovska et al. 2017; Zeng et al. 2021).

A major complication is that the published literature does not represent a single experimental system. Human-skin permeability values are generated across laboratories, diffusion-cell configurations, skin sources and anatomical sites, tissue-preparation procedures, donor and receptor media, temperatures, analytical methods, and reporting conventions. Interstudy variation has therefore been recognized for decades. Fujiwara et al. modeled a latent common permeability underlying study-specific measurements (Fujiwara et al. 2003). Abraham and Martins standardized literature values for temperature and ionization before fitting linear free-energy relationships (Abraham and Martins 2004). Samaras et al. showed that experimental and formulation conditions materially influence in-vitro human skin permeation (Samaras et al. 2012). More recently, HuSkinDB and SkinPiX have made measurement-level permeability records and protocol metadata available in structured public resources (Stepanov et al. 2020; Chedik et al. 2024).

Recent predictive studies further narrow the remaining methodological gap. QSPR models have been developed from curated HuSkinDB records (Waters and Quah 2022), and newer work has combined HuSkinDB and SkinPiX into larger curated resources for predictive modeling (Asgarkhanova et al. 2026). Oh and Ahn explicitly examined disagreement among repeated permeability measurements, protocol variables, record-level versus compound-level modeling, and mixed-effects formulations (Oh and Ahn 2026). Consequently, novelty can no longer rest on the general observations that repeated measurements disagree, that experimental conditions matter, or that mixed-effects models can be fitted to permeability data.

The more operational question is whether source-associated heterogeneity can be **estimated for a previously unseen experimental source and then used to improve predictions for new compounds measured by that source**. If measurements can be separated into a transferable molecular component and a source-associated component shared across compounds, then compounds measured both in a reference corpus and on a new platform could serve as calibration anchors. Under this interpretation, study heterogeneity becomes not only a source of uncertainty but also a parameter that can be estimated when deploying a model to a new laboratory, apparatus, or protocol.

We therefore considered the measurement-level model

$$y_{ijr} = \log_{10} K_{p,ijr} = f(X_i) + \beta^\top Z_{ijr} + b_i + u_j + \epsilon_{ijr},$$

where $K_{p,ijr}$ is the permeability coefficient, expressed in cm/h, for measurement r of compound i reported by source j. Thus, y_{ijr} is the corresponding $\log_{10} K_p$ endpoint. The index r distinguishes repeated measurements of the same compound within the same literature source. $f(X_i)$ is the systematic molecular component, Z_{ijr} contains measurement-level experimental covariates, b_i is residual compound-specific variation not explained by $f(X_i)$, u_j is a source-associated random intercept, and ϵ_{ijr} is measurement-level residual error. For a previously measured compound, the total chemical contribution is $f(X_i) + b_i$; for a genuinely unseen compound, only $f(X_i)$ is available in expectation.

The analysis was designed as development, replication in a separately curated database, and cross-dataset transfer. The Cheruvu et al. dataset (Cheruvu et al. 2022) was used to establish the variance structure and compare conventional compound aggregation with measurement-level hierarchical prediction. HuSkinDB was then used to replicate the hierarchy and to simulate deployment by holding out entire literature sources. Finally, SkinPiX was kept separate as an external transfer set: the molecular model was trained on HuSkinDB only, shared HuSkin–SkinPiX compounds were used as anchors, and SkinPiX-only compounds were reserved for evaluation. The central hypotheses were that (i) permeability measurements contain a reproducible source-associated component not captured by molecular descriptors alone, and (ii) this component can be estimated from a small number of shared compounds to improve prediction at an unseen source.

3 Materials and Methods

3.1 Study design and data sources

Three public human-skin permeability datasets were analyzed sequentially (Table 1). The Cheruvu et al. 2022 compilation was used for model development (Cheruvu et al. 2022) and contained 476 retained measurements from 148 compounds and 69 literature references. HuSkinDB was used for replication of the hierarchical findings and for internal source-calibration experiments (Stepanov et al. 2020); the processed dataset contained 546 measurements from 251 compounds and 95 references. SkinPiX was used only for cross-dataset transfer (Chedik et al. 2024), with 199 retained measurements from 108 compounds and 36 references after preprocessing. Dataset-specific exclusions and provenance audits are reported in the supplementary material.

Table 1: Dataset structure and role in the analysis.

Dataset	Measurements	Compounds	Sources	Repeated compounds	Multi-source compounds	Role
Cheruvu	476	148	69	81	46	Development
HuSkinDB	546	251	95	89	61	Replication and held-out-source calibration
SkinPiX	199	108	36	65	14	Cross-dataset transfer

The datasets were deliberately not pooled before validation. Their separation allowed development, database-level replication, and cross-dataset transfer to be evaluated sequentially rather than within a single merged corpus. HuSkinDB was treated as a separately curated replication dataset rather than assumed to be statistically independent of Cheruvu at the underlying literature-source level. For the HuSkinDB-to-SkinPiX transfer analysis, literature-source overlap was assessed explicitly by DOI as described below.

3.2 Harmonization and compound/source identities

Throughout the analysis, the response endpoint was $\log_{10} K_p$, with K_p expressed in cm/h. Dataset-specific permeability values were harmonized to this scale before modeling.

In the Cheruvu dataset, canonical compound identity was based primarily on normalized compound name rather than CAS number because the source table contained at least one CAS collision between chemically distinct entries. Reference identity was taken from the literature-source field. The supplied `logkpl` endpoint was already expressed as $\log_{10}$ permeability with K_p in cm/h and was retained on that scale. Experimental temperature values reported in Kelvin were converted to degrees Celsius.

HuSkinDB compound identity was defined using SMILES where available, with normalized compound names used as fallback. Literature sources were identified preferentially by DOI and otherwise by the database reference field. HuSkinDB reports $\log_{10} K_p$ with K_p in cm/s; values were converted to the common cm/h scale by adding $\log_{10}(3600)$.

For SkinPiX, compound identity was defined from SMILES with normalized names as fallback. Source identity was based on publication identifier and DOI. Exact equality measurements were retained; censored inequality records were excluded rather than treated as exact K_p values. Cross-database HuSkin–SkinPiX compound overlap was identified first by exact SMILES and then by unambiguous normalized-name matching. All resulting overlap assignments were manually audited. No DOI overlap was identified between the HuSkin and SkinPiX sources used for transfer.

Because HuSkinDB and SkinPiX did not provide identical molecular-descriptor definitions, molecular weight and Crippen logP were recomputed from SMILES using RDKit for **both** databases (Landrum et al. 2026). All HuSkin and SkinPiX structures parsed successfully. Descriptor concordance among matched compounds

was verified in the transfer audit, supporting compatibility of the recomputed molecular descriptors across databases.

3.3 Hierarchical decomposition in the development dataset

The initial Cheruvu hierarchy was examined using nested linear mixed-effects models fitted with `lme4` (Bates et al. 2015):

$$
\begin{aligned}
M_0 &: y_{ijr} = \alpha + b_i + \epsilon_{ijr}, \\
M_1 &: y_{ijr} = \alpha + b_i + u_j + \epsilon_{ijr}, \\
M_2 &: y_{ijr} = \alpha + \beta_T T_{ijr} + b_i + u_j + \epsilon_{ijr}, \\
M_3 &: y_{ijr} = \alpha + \beta_T T_{ijr} + \beta_H H_{ijr} + b_i + u_j + \epsilon_{ijr},
\end{aligned}
$$

where T_{ijr} is donor/skin-surface temperature and H_{ijr} is skin thickness for measurement r of compound i in source j.

Compound and reference were modeled as crossed random intercepts. Maximum-likelihood fits were used to compare fixed-effect specifications; restricted maximum likelihood (REML) fits were used for final parameter and variance-component estimation. For presentation, the compound, source, and residual variance components were expressed as proportions of their sum, $\sigma_b^2 + \sigma_u^2 + \sigma_\epsilon^2$. These percentages therefore describe the partitioning of modeled random/residual variance and do not include variation attributable to fixed effects. Because testing a variance component against zero places the null hypothesis on the parameter boundary, the addition of the reference random effect was interpreted primarily using the large change in information criteria and the estimated variance component rather than relying on a naive chi-square test. Temperature and skin-thickness additions were compared using ordinary likelihood-ratio tests.

3.4 Molecular QSPR specifications

The molecular component $f(X)$ was prespecified rather than selected by stepwise search. Two specifications were evaluated in Cheruvu:

S1: size–lipophilicity model

$$f_{S1}(X_i) = \beta_0 + \beta_1 MW_i + \beta_2 \log K_{ow,i}.$$

S2: extended physicochemical model

$$f_{S2}(X_i) = \beta_0 + \beta_1 MW_i + \beta_2 \log K_{ow,i} + \beta_3 MP_i + \beta_4 \log S_{aq,i} + \beta_5 HBD_i + \beta_6 HBA_i.$$

Descriptors were standardized using compound-level training-set means and standard deviations. The molecular terms were fitted jointly with the selected hierarchical structure rather than estimated from pre-aggregated permeability values. The explanatory contribution of $f(X)$ was summarized as the proportional reduction in compound random-effect variance relative to the hierarchy-only model.

For conventional QSPR comparison, measurements were collapsed to a mean log permeability per compound and molecular descriptors were summarized by their compound median. This provided a conventional compound-level benchmark while keeping the molecular information matched as closely as possible to the hierarchical models.

HuSkinDB replication used S1 only because molecular weight and logP were complete for all records, whereas donor temperature was missing in 155 of 546 measurements and the broader Cheruvu descriptor panel was not available in directly comparable form. Temperature was therefore evaluated as a secondary complete-case analysis in HuSkinDB rather than as a required term in the primary HuSkin QSPR.

Accordingly, variance decompositions from Cheruvu et al. and HuSkinDB were used to assess replication of a substantial source-associated component, not to compare the absolute magnitude of variance components directly between datasets. The Cheruvu decomposition adjusts for temperature as a fixed effect, whereas the primary HuSkinDB decomposition does not.

3.5 Leave-one-compound-out validation and prediction estimands

Prediction was evaluated using leave-one-compound-out (LOCO) validation. In each fold, all measurements of one compound were removed before model fitting and descriptor standardization. Three estimands were distinguished explicitly:

1. **Conventional aggregate QSPR:** prediction from the compound-level regression.
2. **Hierarchical, source-blind:** prediction from the hierarchical molecular fixed effects with all random effects set to zero for the held-out compound.
3. **Hierarchical, source-aware:** prediction using the molecular fixed effects plus the estimated source random effect when the held-out measurement belonged to a source that remained represented in the training fold. If removal of the held-out compound also removed that source entirely from training, the source effect was unavailable and was set to zero.

This distinction was critical because the source-aware hierarchical predictor could use source-specific information when that source remained represented in the training fold, whereas the conventional aggregate model could not. The source-blind prediction therefore served as a direct test of whether hierarchical estimation itself produced a superior transferable molecular function $f(X)$, whereas the source-aware prediction quantified the additional value of source context available from the training data.

Performance was summarized using RMSE, MAE, predictive R^2, and compound-balanced RMSE. Sensitivity analyses in Cheruvu examined exclusion of observations with absolute standardized residuals greater than three, descriptor aggregation by mean rather than median, restriction to multi-reference compounds, and an expanded nonlinear molecular specification. These analyses were secondary and did not alter the prespecified primary specifications. Unless otherwise stated, headline LOCO RMSE was calculated across all held-out measurement records, so compounds contributing multiple retained measurements contributed multiple residuals to the pooled error estimate. Compound-balanced RMSE, in which compounds contributed equally after within-compound aggregation, was reported as a secondary sensitivity metric.

3.6 Chemical applicability domain

Chemical applicability was assessed using descriptors only, separately from experimental-source applicability. For every LOCO fold, descriptor means, standard deviations, covariance matrices, and univariate ranges were estimated **from the training compounds only**. A held-out compound was considered in-domain when (i) its squared Mahalanobis distance was below the 95th percentile of the appropriate chi-square distribution and (ii) every descriptor remained within the corresponding univariate training range. S1 therefore used a two-dimensional domain and S2 a six-dimensional domain.

For the SkinPiX transfer analysis, applicability was defined relative to the complete HuSkin training descriptor space because HuSkinDB was the deployed reference model and SkinPiX was external. No SkinPiX information was used to define the chemical domain.

3.7 Held-out-source calibration in HuSkinDB

To simulate deployment to an unseen experimental source, each HuSkin reference was held out in turn. The model was fitted to all other references. Compounds present both in the held-out source and in the training

data were considered candidate **anchor compounds**. Compounds present in the held-out source but absent from all model-fitting data were reserved as genuinely unseen test compounds.

For an anchor compound i measured in held-out source L, the residual for measurement r relative to the fitted reference model was

$$e_{iLr} = y_{iLr} - \hat{\mu}_{iLr},$$

where $\hat{\mu}_{iLr}$ included the molecular fixed effect and the known compound-specific random effect estimated from the training sources, but no effect for the held-out source. If an anchor compound had multiple measurements within source L, its residual contribution was first averaged within compound,

$$\bar{e}_{iL} = \frac{1}{n_{iL}} \sum_{r=1}^{n_{iL}} e_{iLr},$$

where n_{iL} is the number of measurements of compound i in source L.

For a selected panel $A_L^{(k)}$ containing k anchor compounds, the raw source offset was then

$$\bar{e}_L^{(k)} = \frac{1}{k} \sum_{i \in A_L^{(k)}} \bar{e}_{iL}.$$

The estimated source effect was shrunk toward zero using

$$\hat{u}_L^{(k)} = \frac{k\hat{\sigma}_u^2}{\hat{\sigma}_\epsilon^2 + k\hat{\sigma}_u^2} \bar{e}_L^{(k)}.$$

This shrinkage rule was used as a pragmatic empirical-Bayes approximation in which each anchor compound contributed equally after repeated measurements had been averaged within compound. Thus, k denotes the number of chemically distinct anchor compounds rather than the total number of permeability measurements. The approximation does not explicitly account for differences in precision caused by unequal numbers of repeated measurements among anchors or for uncertainty in the training-derived compound random effects.

Predictions for unseen compounds in the held-out source were then shifted by $\hat{u}_L$. Anchor-panel sizes from 1 to 5 compounds were evaluated where available. For each panel size, 100 random anchor selections were drawn with a fixed random seed.

Prediction error was first calculated separately within each held-out source. For source s and anchor-selection replicate q, RMSE was calculated across the unseen test measurements from that source. For each anchor-panel size, these source-specific RMSE values were averaged across the 100 anchor selections and then across eligible sources, giving equal weight to each experimental source regardless of the number of test measurements it contributed. We refer to this quantity as the mean source-level RMSE. Percentage improvement from calibration was calculated as

$$100 \left(1 - \frac{\overline{\mathrm{RMSE}}_{\mathrm{calibrated}}}{\overline{\mathrm{RMSE}}_{\mathrm{uncalibrated}}} \right).$$

Results were summarized both across all sources eligible for a given panel size and in matched source cohorts supporting the same panel sizes. Because the set of eligible sources changes as the required number of anchors increases, inference about anchor number was based primarily on matched cohorts. The 9-source cohort supporting all panel sizes from 1 to 5 anchors was used for the primary HuSkin anchor-number comparison.

3.8 Falsification by wrong-source calibration

A negative-control analysis tested whether calibration improvement could arise from adding a generic offset rather than information specific to the target experimental source. To make the comparison directly controlled, the falsification experiment used a pairwise double-holdout design.

For each ordered pair consisting of a target source L and a different donor source D, both sources were excluded before fitting the reference model. Thus, the matched-source and different-source offsets for a given target–donor pair were estimated against the same molecular coefficients, descriptor standardization, compound random effects, source and residual variance estimates, and empirical-Bayes shrinkage rule.

Target test compounds were defined using the ordinary leave-target-source-out criterion: compounds measured in L that were absent from all other HuSkin sources. This test set was fixed before selecting donor source D, so compounds did not become artificially novel merely because the donor source was additionally excluded. Candidate anchors for L and D were then defined relative to the common double-holdout training corpus, and a pair was eligible for anchor-panel size k only when both sources had at least k compounds represented in that training set.

For each eligible target–donor pair and anchor-panel size k, 100 independently sampled panels of k anchor compounds were drawn from the target source and from the donor source. Repeated measurements of an anchor compound were first averaged within compound as in the primary source-calibration analysis. The corresponding empirical-Bayes offsets were

$$\hat{u}_L^{(k)} = \frac{k\hat{\sigma}_u^2}{\hat{\sigma}_\epsilon^2 + k\hat{\sigma}_u^2}\bar{e}_L^{(k)}$$

and

$$\hat{u}_D^{(k)} = \frac{k\hat{\sigma}_u^2}{\hat{\sigma}_\epsilon^2 + k\hat{\sigma}_u^2}\bar{e}_D^{(k)}.$$

The same uncalibrated predictions for the novel test measurements in target source L were then evaluated after adding either the matched offset $\hat{u}_L^{(k)}$ or the different-source offset $\hat{u}_D^{(k)}$. RMSE and MAE were averaged across anchor-selection replicates within each target–donor pair.

Because a target source could have multiple eligible donor sources, pair-level results were next averaged across donor sources within each target source. The resulting target-source summaries were then averaged across eligible target sources, giving each target source equal weight regardless of the number of eligible donors or novel test measurements it contributed. Comparisons were made within anchor-panel size because the eligible target-source cohort changed as k increased.

The primary falsification contrast was

$$\Delta_k = \overline{\mathrm{RMSE}}_{\text{different source}} - \overline{\mathrm{RMSE}}_{\text{matched}},$$

where positive values indicate that calibration using information from a different source produced greater prediction error than calibration using the target source itself.

3.9 Cross-dataset transfer from HuSkinDB to SkinPiX

The SkinPiX analysis was designed as a stricter cross-dataset transfer test. A hierarchical S1 model was fitted to HuSkinDB using RDKit molecular weight and logP. SkinPiX was not used for model fitting. Within each SkinPiX source, compounds also present in HuSkinDB were candidate anchors, while SkinPiX-only compounds were the test set. A source was eligible when it contained at least one anchor, at least one SkinPiX-only compound, and no DOI overlap with HuSkinDB.

Cross-dataset calibration performance was summarized using the same source-balanced procedure as in the HuSkin held-out-source experiment. RMSE was calculated separately for the SkinPiX-only test measurements within each source and each anchor-selection replicate, averaged across the 100 anchor selections, and then averaged across sources. Thus, each SkinPiX source contributed equally to the reported mean source-level RMSE regardless of its number of test measurements.

The same empirical-Bayes calibration procedure used in the HuSkin internal experiment was then applied to each eligible SkinPiX source. Anchor panels were evaluated where possible with 100 random anchor selections. Because only 4 SkinPiX sources simultaneously contained anchors and genuinely novel compounds, the one-anchor result across these sources was treated as the primary external-transfer result. A matched 3-source subset supporting up to 3 anchors was used only as a secondary anchor-number sensitivity analysis. Larger anchor panels involved progressively fewer sources and were treated descriptively rather than as evidence of a general dose-response.

For compound-level visualization of the one-anchor experiment, calibrated predictions were averaged across the 100 random one-anchor selections for each measurement. This visualization therefore represents expected performance over random selection of one available anchor rather than a single privileged anchor choice.

3.10 Uncertainty estimation

Uncertainty was estimated at the level corresponding to the independent validation unit. For leave-one-compound-out analyses, 95% confidence intervals were obtained by paired cluster bootstrap of held-out compounds. Compounds were sampled with replacement, with all measurements from each selected compound retained together, and RMSE was recalculated from the existing out-of-fold predictions. Within each bootstrap replicate, RMSE was recalculated across the retained measurement records, preserving the measurement-weighted primary LOCO estimand. The same bootstrap sample was used for all prediction approaches within each replicate, preserving the paired comparison. Relative RMSE improvement was calculated against the conventional aggregate QSPR within each bootstrap sample. 2000 bootstrap replicates were used, with percentile intervals defined by the 2.5th and 97.5th percentiles.

Uncertainty in variance partitioning was estimated by parametric bootstrap of the final REML mixed-effects models. For each simulated dataset, the model was refitted and the compound, source, and residual variance components were normalized by their sum. Ninety-five percent percentile intervals were calculated from 2000 bootstrap replicates.

For the matched 9-source HuSkin calibration analysis, uncertainty in mean source-level RMSE improvement was estimated by nonparametric bootstrap of literature sources. 9 sources were sampled with replacement in each replicate, with the same sampled source set retained across anchor-panel sizes. Improvement was recalculated using the same ratio of mean calibrated to mean uncalibrated source-level RMSE defined above.

For the pairwise double-holdout falsification analysis, uncertainty was estimated by nonparametric bootstrap of target literature sources within each anchor-panel size. The target-source summaries had already averaged across eligible donor sources, so target sources rather than target–donor pairs were treated as the independent resampling units. Matched-source and different-source calibration were evaluated on the same resampled target sources, and the paired difference in mean source-level RMSE was recalculated in each bootstrap replicate.

Formal confidence intervals were not calculated for the external SkinPiX result because only 4 eligible sources were available. External-transfer heterogeneity was instead summarized using individual source results and the number of sources and compounds showing lower error after calibration.

3.11 Software and reproducibility

All statistical analyses were performed in R. Linear mixed-effects models were fitted with `lme4` (Bates et al. 2015). RDKit was used in a separate Python environment to calculate molecular weight and Crippen logP

from SMILES (Landrum et al. 2026). Analysis scripts were organized sequentially so that data preparation, model fitting, validation, falsification, and manuscript output were separated and reproducible. No results were recalculated in the manuscript-generation scripts.

4 Results

4.1 Source-associated heterogeneity replicates across datasets

Adding a literature-source random intercept to the Cheruvu hierarchy reduced AIC from 1033.64 to 823.29. Adding temperature further reduced AIC to 812.32, whereas adding skin thickness increased AIC slightly to 813.66. The temperature coefficient was 0.0425 $\log_{10}$ units per °C ($p =< 0.001$), corresponding to approximately a 10.3% multiplicative increase in K_p per degree Celsius. Skin thickness was not supported ($p = 0.42$). The M2 model was therefore used as the primary Cheruvu hierarchy.

In the REML variance decomposition, the modeled random/residual variance in Cheruvu et al. was partitioned into compound-associated (66.4%, 95% CI 56.9–74.7%), source-associated (22.8%, 14.4–31.9%), and residual (10.8%, 8.3–14.0%) components (Table 2; Figure 1). HuSkinDB reproduced substantial source-associated variation: compound, source, and residual components accounted for 23.5% (15.2–32.9%), 44.4% (33.5–54.5%), and 32.1% (24.7–40.8%), respectively. Because the Cheruvu model included temperature as a fixed effect whereas the primary HuSkinDB model did not, these percentages should not be interpreted as a direct quantitative comparison between datasets. Rather, the replicated result is that a substantial source-associated component was present in both datasets.

Table 2: Partitioning of modeled random/residual variance. Percentages are normalized over the compound, source, and residual variance components; fixed-effect variation is not included in the denominator. The Cheruvu and HuSkinDB decompositions use different fixed-effect specifications and should therefore be interpreted as replication of substantial source-associated variation in a separately curated database rather than compared directly in magnitude.

Dataset	Variance component	Estimate (%)	95% CI
Cheruvu	Compound	66.4	[56.9, 74.7]
Cheruvu	Source	22.8	[14.4, 31.9]
Cheruvu	Residual	10.8	[8.3, 14.0]
HuSkinDB	Compound	23.5	[15.2, 32.9]
HuSkinDB	Source	44.4	[33.5, 54.5]
HuSkinDB	Residual	32.1	[24.7, 40.8]

4.2 Molecular descriptors explain substantial compound variation, but source context drives most of the predictive gain

In Cheruvu, S1 (molecular weight plus logKow) explained 60.3% of the compound random-effect variance. The broader S2 specification explained 61.4%, only 1.1 percentage points more. Thus, most of the explainable compound signal captured by these prespecified models was already present in the simple size–lipophilicity specification. In HuSkinDB, molecular weight plus logP explained 37.1% of between-compound variance.

LOCO validation separated improvement in the molecular component from improvement due to source context. Relative to conventional aggregation, source-blind hierarchical prediction changed RMSE by 5.4% (95% CI

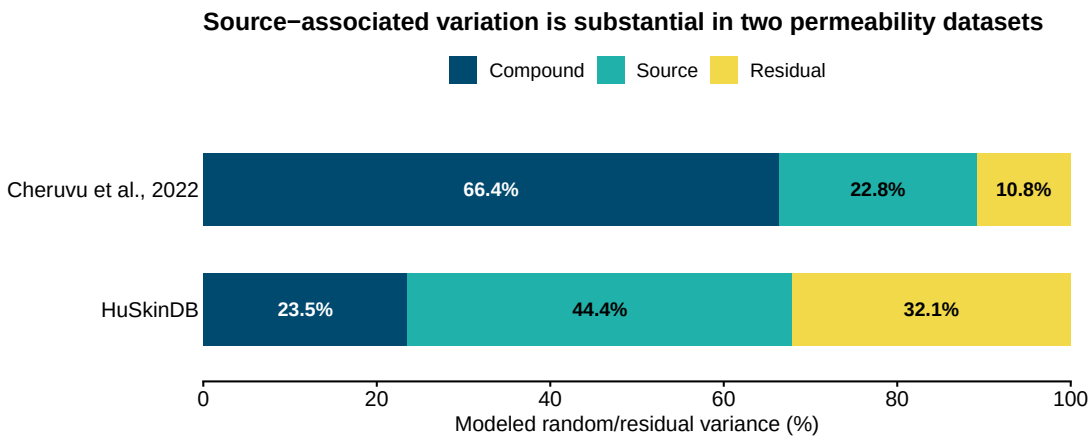


Figure 1: Variance partitioning in Cheruvu et al. 2022 and HuSkinDB. Bars show the relative contributions of compound, literature source, and residual components to modeled random/residual variance. The Cheruvu decomposition adjusts for temperature, whereas the primary HuSkinDB decomposition does not because temperature was incomplete; the figure therefore demonstrates replication of substantial source-associated variation rather than a direct comparison of component magnitudes between datasets.

-3.6–10.9%) for Cheruvu S1, 7.0% (-5.1–14.0%) for Cheruvu S2, and -1.7% (-5.1–1.0%) for HuSkinDB S1. All three intervals included zero, providing no evidence that hierarchical fitting alone consistently improved the transferable molecular predictor.

In contrast, source-aware hierarchical prediction improved RMSE by 16.7% (95% CI 11.6–21.8%) for Cheruvu S1, 18.7% (11.5–23.3%) for Cheruvu S2, and 17.0% (10.0–23.8%) for HuSkinDB S1. The intervals excluded zero in all three analyses, showing that the reproducible predictive benefit arose primarily when source-specific information was available (Table 3; Figure 2).

Table 3: Leave-one-compound-out prediction under conventional aggregate, source-blind hierarchical, and source-aware hierarchical settings. Source-aware prediction used an estimated source effect when the source remained represented in the training fold; otherwise the source effect was set to zero.

Dataset	Specification	Prediction setting	RMSE (95% CI)	RMSE improvement vs conventional (%)
Cheruvu	S1	Conventional aggregate QSPR	0.866 [0.725, 1.017]	Reference
Cheruvu	S1	Hierarchical, source blind	0.819 [0.708, 0.945]	5.4 [-3.6, 10.9]
Cheruvu	S1	Hierarchical, source aware	0.721 [0.601, 0.855]	16.7 [11.6, 21.8]
Cheruvu	S2	Conventional aggregate QSPR	0.850 [0.715, 0.977]	Reference
Cheruvu	S2	Hierarchical, source blind	0.790 [0.703, 0.883]	7.0 [-5.1, 14.0]
Cheruvu	S2	Hierarchical, source aware	0.691 [0.602, 0.788]	18.7 [11.5, 23.3]
HuSkinDB	S1	Conventional aggregate QSPR	1.333 [1.198, 1.474]	Reference
HuSkinDB	S1	Hierarchical, source blind	1.355 [1.201, 1.537]	-1.7 [-5.1, 1.0]
HuSkinDB	S1	Hierarchical, source aware	1.106 [0.959, 1.305]	17.0 [10.0, 23.8]

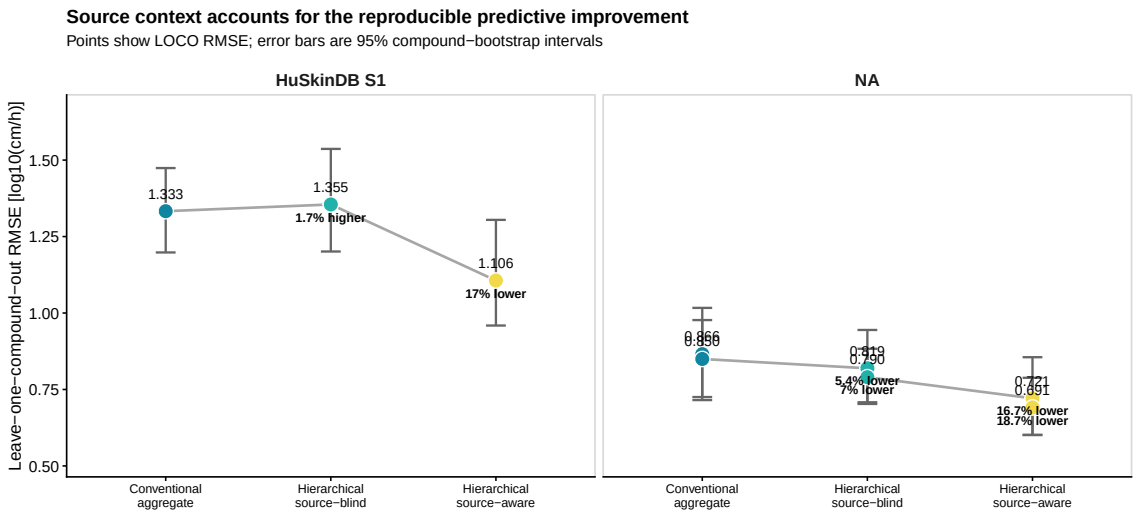


Figure 2: Leave-one-compound-out RMSE under conventional aggregate QSPR, source-blind hierarchical prediction, and source-aware hierarchical prediction. Source-aware prediction incorporated an estimated source random effect when the source remained represented in the training fold; otherwise the source effect was zero. Error bars show 95% compound-bootstrap intervals. Source-blind effects were small and uncertain, whereas access to source context produced the larger and reproducible reduction in prediction error.

4.3 Shared compounds calibrate completely held-out HuSkin sources

Holding out entire HuSkin references tested whether source information could be recovered rather than assumed known. Calibration used only compounds shared between the held-out source and the training corpus; evaluation was restricted to compounds from that source that were absent from all model-fitting data.

Calibration improved mean source-level RMSE at every tested anchor-panel size among eligible sources. Because eligibility changed with panel size, the primary anchor-number comparison used the same 9 references that supported 1 through 5 anchors. Within this matched cohort, mean source-level RMSE improvement was 31.1% (95% CI 13.4–42.1%) with 1 anchor and 45.3% (19.2–60.0%) with 5 anchors. Point estimates increased monotonically with anchor-panel size, although the overlapping intervals do not support interpreting each incremental increase as statistically distinct.

The cohort-level trend did not imply universal source-level success. At 5 anchors, 8 of 9 matched sources improved, while HUR_48 deteriorated substantially. The median source-specific RMSE improvement at 5 anchors was 40.0%, whereas the reduction in mean source-level RMSE was 45.3%. These quantities differ because the latter is calculated from the ratio of the mean calibrated and uncalibrated source-level RMSEs, rather than as the arithmetic mean or median of source-specific percentage improvements.

Table 4: Held-out HuSkin source calibration in the matched-source cohort. Confidence intervals were obtained by bootstrap resampling of literature sources.

Anchors	Matched sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)	95% CI
1	9	1.135	0.782	31.1	[13.4, 42.1]
2	9	1.135	0.705	37.9	[16.0, 51.0]
3	9	1.135	0.675	40.5	[17.2, 54.8]
4	9	1.135	0.641	43.6	[18.4, 57.8]

Anchors	Matched sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)	95% CI
5	9	1.135	0.621	45.3	[19.2, 60.0]

4.4 Wrong-source calibration falsifies a generic-offset explanation

The stricter pairwise double-holdout control reproduced the same qualitative pattern as the primary calibration analysis. For each target–donor pair, both offsets were estimated against the same reference model fitted after excluding both sources, so the molecular model, scaling, variance estimates, shrinkage rule, and target test set were common to the matched- and different-source predictions.

At one anchor, matched-source calibration changed mean source-level RMSE by 8.6%, whereas calibration using an offset from a different source changed RMSE by -8.6%. At five anchors, the corresponding values were 45.1% and -11.4%, respectively (Table 5; Figure 4). Because the eligible target-source cohort changed with anchor-panel size, these marginal percentages were interpreted within rather than across panel sizes.

More importantly, the direct paired contrast between different-source and matched-source RMSE was positive at every anchor-panel size. The difference was 0.212 $\log_{10}$(cm/h) (95% CI [0.112, 0.324]) with one anchor and 0.645 ([0.406, 0.941]) with five anchors, with the intermediate intervals also remaining above zero. Thus, an offset estimated from the target source consistently outperformed an equivalently estimated offset derived from another held-out source.

Because both offsets within each target–donor comparison were estimated against the same double-holdout reference model, this result is not attributable to differences between separately fitted reference models. It instead supports the interpretation that the useful calibration signal is specifically associated with the experimental source containing the target measurements.

Table 5: Pairwise double-holdout falsification comparing calibration from the target source with calibration from a different HuSkin source. For each target–donor pair, both offsets were estimated against the same reference model with both sources excluded. Positive values of the RMSE contrast indicate higher prediction error under different-source calibration.

Anchors	Target sources	Matched-source RMSE improvement (%)	95% CI	Different-source RMSE improvement (%)	95% CI	Different-source - matched RMSE	95% CI
1	31	8.6	[-1.2, 17.8]	-8.6	[-14.2, -4.9]	0.212	[0.112, 0.324]
2	22	15.6	[2.0, 28.8]	-11.3	[-19.8, -5.9]	0.313	[0.186, 0.478]
3	19	20.9	[0.9, 37.4]	-14.2	[-24.7, -6.6]	0.352	[0.189, 0.569]
4	12	36.1	[14.4, 51.8]	-13.5	[-25.9, -4.3]	0.545	[0.337, 0.818]
5	9	45.1	[20.4, 59.8]	-11.4	[-32.3, 0.6]	0.645	[0.406, 0.941]

4.5 HuSkin-trained calibration transfers to SkinPiX sources

The cross-dataset analysis tested the same mechanism under a stricter separation between model development and deployment. The molecular model was fitted to 546 HuSkin measurements from 251 compounds. SkinPiX contained compounds overlapping HuSkinDB and SkinPiX-only compounds as documented in the transfer audit. Of the sources containing HuSkin-overlapping compounds, 4 contained both an available anchor and at least one SkinPiX-only compound suitable for genuinely external evaluation. Across these sources, the test set comprised 54 unique compounds not present in HuSkinDB.

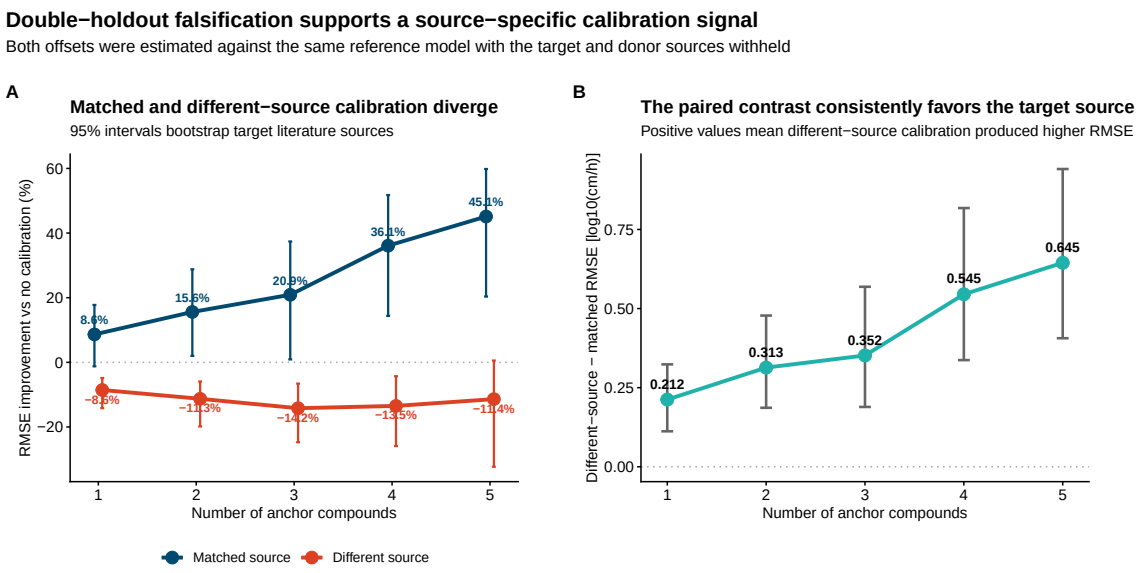


Figure 3: Pairwise double-holdout falsification of a generic-offset explanation. For each target–donor source pair, both calibration offsets were estimated against the same reference model after excluding both sources. Panel A shows mean source-level RMSE improvement for matched-source and different-source calibration, with 95% target-source bootstrap intervals. Panel B shows the direct paired difference between different-source and matched-source RMSE; positive values indicate poorer prediction when the offset is borrowed from another source. All paired confidence intervals remained above zero.

With one randomly selected anchor compound per source, calibration reduced mean source-level RMSE by 17.6% across the 4 eligible SkinPiX sources. Because RMSE was calculated within source and then averaged across sources, each source contributed equally to this estimate regardless of its number of test measurements. At the source level, 3 of 4 sources improved. At the compound level, 44 of 54 unseen SkinPiX compounds had lower mean absolute error after calibration when predictions were averaged across the 100 one-anchor selections. The source-level RMSE and compound-level absolute-error summaries therefore describe complementary but distinct aspects of transfer performance.

3 of the 4 eligible SkinPiX sources supported at least 3 anchors. Restricting the analysis to this matched subset, mean source-level RMSE improvement increased across the supported anchor-panel sizes, from 18.6% at the smallest panel to 23.9% at the largest. This small matched subset provides supportive rather than definitive evidence for an anchor-number effect across databases. Estimates based on still larger anchor requirements involved progressively fewer sources and were not emphasized.

All 54 external SkinPiX test compounds were inside the HuSkin-defined chemical applicability domain. The transfer result should therefore be interpreted primarily as **experimental-source transfer within HuSkin chemical space**, not as evidence for reliable chemical extrapolation beyond that space.

Table 6: Cross-dataset calibration of SkinPiX sources using a model trained only on HuSkinDB.

Cohort	Anchors	Sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)
All eligible sources	1	4	0.810	0.667	17.6
Matched three-source cohort	1	3	0.949	0.772	18.6
Matched three-source cohort	2	3	0.949	0.740	22.1
Matched three-source cohort	3	3	0.949	0.722	23.9

4.6 Applicability-domain diagnostics

Applicability-domain analysis was used as a deployment qualification rather than as an additional model-selection criterion. Cheruvu and HuSkinDB applicability-domain counts are reported in Supplementary Table S7. All 54 SkinPiX compounds used in the external transfer test were in-domain relative to the HuSkin

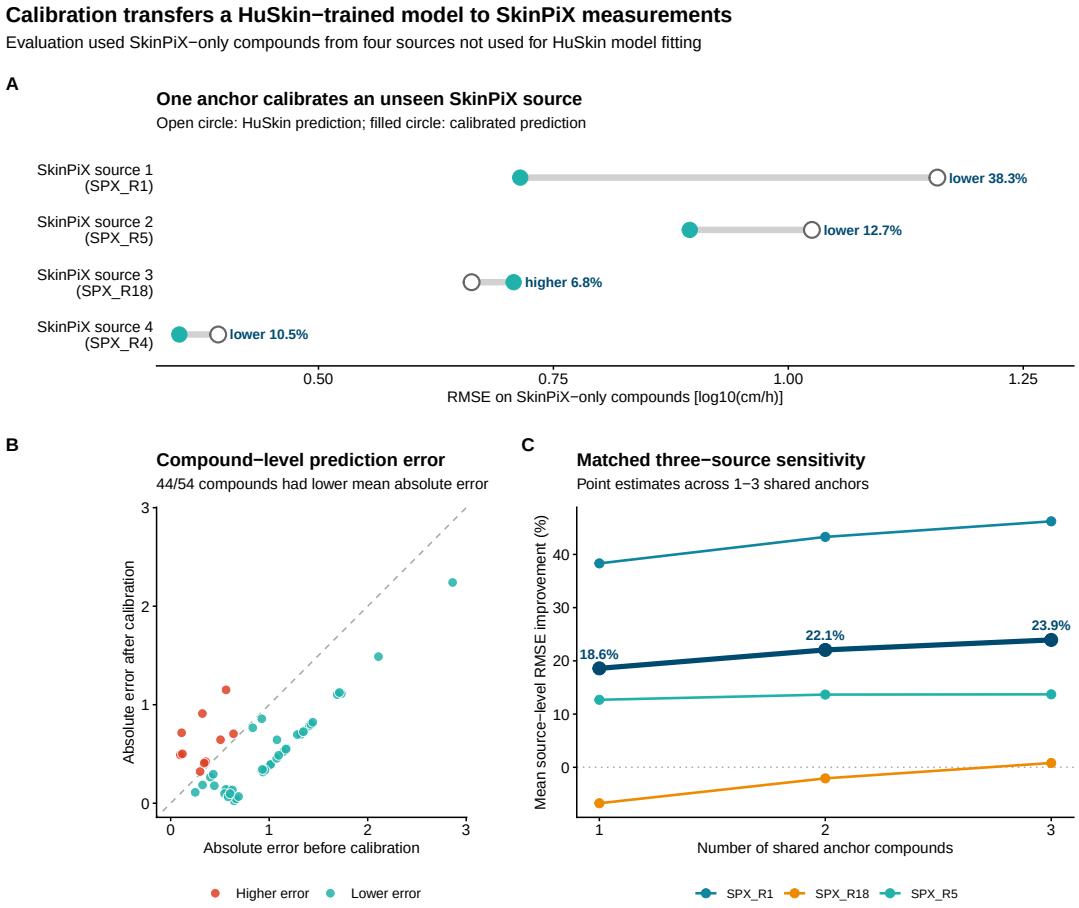


Figure 4: Cross-dataset transfer from HuSkinDB to SkinPiX. Panel A compares uncalibrated and one-anchor calibrated RMSE within each eligible unseen SkinPiX source. Panel B compares compound-level absolute prediction error before and after calibration for SkinPiX-only compounds. Panel C shows the matched source subset supporting multiple anchors; colored lines represent individual sources and the dark line represents reduction in mean source-level RMSE across those same sources.

that heterogeneity behaves like a **calibratable source-level quantity** that can be estimated from shared compounds and then transferred to compounds not used in fitting.

5.2 Hierarchical fitting does not automatically create a better universal molecular QSPR

The source-blind analysis narrows the interpretation of the LOCO results. Although the point estimates in Cheruvu suggested lower RMSE under source-blind hierarchical fitting, the corresponding bootstrap intervals included no improvement. HuSkinDB similarly showed no source-blind benefit. In contrast, the larger reductions obtained when source context was available had confidence intervals entirely above zero in both development and replication datasets. The evidence therefore does not support hierarchical fitting as a consistently superior estimator of the universal molecular function $f(X)$; the reproducible predictive gain arises from representing source-specific information.

This is scientifically useful because it argues against framing the method as a universally superior estimator of $f(X)$. The data instead support a two-part prediction problem. Molecular descriptors determine the expected permeability of an unseen compound, while source information shifts that expectation toward the scale of a particular experimental system. The value of the hierarchical formulation is therefore not that it always discovers a better molecular equation, but that it separates molecular prediction from source calibration.

The modest difference between Cheruvu S1 and S2 reinforces this point. Expanding from molecular weight and logKow to the broader S2 descriptor set increased explained compound variance by only 1.1 percentage points. A more complex descriptor search might improve point prediction, but it would obscure the methodological question addressed here and increase the risk that performance differences were driven by feature selection rather than by the treatment of measurement hierarchy.

5.3 Anchor compounds convert an unseen source from unknown to calibratable

The held-out-source experiment is closest to the intended deployment setting. A new source initially has no estimated random effect, so prediction defaults to the reference scale. Once shared compounds are measured, their residuals relative to the reference model provide information about the new source's systematic displacement. Empirical-Bayes shrinkage prevents a small anchor panel from being interpreted as a perfectly known offset.

The matched 9-source HuSkin analysis showed a monotonic point-estimate improvement as anchors increased from 1 to 5. Bootstrap intervals remained above zero at every anchor-panel size, including 31.1% (95% CI 13.4–42.1%) with 1 anchor and 45.3% (19.2–60.0%) with 5 anchors. However, the substantial overlap between panel-size intervals means that the monotonic point-estimate trend should not be interpreted as evidence that each additional anchor confers a statistically distinct benefit. HUR_48 also demonstrates that a single source-wide intercept is not universally adequate. Possible explanations include compound-by-protocol interactions, condition changes within a publication, incorrect metadata grouping, formulation effects, or other deviations from the common-offset assumption. That failure case should be treated as part of the method's operating characteristics rather than removed post hoc.

Most importantly, the pairwise double-holdout falsification supports the proposed source-specific mechanism. Within each target–donor comparison, both offsets were estimated against the same reference model, eliminating differences in molecular coefficients, descriptor scaling, variance estimates, or shrinkage as explanations for the contrast. The direct difference between different-source and matched-source RMSE was positive at every anchor-panel size, with all target-source bootstrap confidence intervals remaining above zero.

If calibration improvement arose primarily from arbitrary recentering, regression toward an overall mean, or incompatibility between separately fitted reference models, an offset estimated from another held-out source under the same model should have provided comparable benefit. Instead, prediction was consistently better when the calibration information came from the source containing the target measurements. The information

recovered from anchor compounds therefore appears to contain a source-associated component shared across compounds within that source.

5.4 Cross-dataset transfer provides the strongest deployment-oriented evidence

SkinPiX was intentionally withheld from model development. HuSkinDB provided the molecular model, SkinPiX-overlapping compounds supplied anchors, and SkinPiX-only compounds supplied the test set. No DOI overlap was identified between the two source sets used for transfer, and RDKit descriptors were recalculated from structure in both databases to avoid descriptor-definition mismatch.

The resulting one-anchor improvement of 17.6% across 4 SkinPiX sources is therefore qualitatively different from an internal cross-validation gain. It shows that a source-calibration procedure developed and validated inside one database can improve predictions in another curated database. The result should nevertheless be interpreted with the correct level of confidence. Only 4 SkinPiX sources were eligible for true anchor-plus-novel-compound testing, and 1 did not improve with one-anchor calibration. The external evidence is therefore promising and mechanistically consistent with the HuSkin experiment, but not yet large enough to establish the frequency with which calibration will succeed in arbitrary laboratories or protocols. Formal confidence intervals were not calculated for this external-transfer estimate because only 4 eligible SkinPiX sources were available; treating the 54 compounds as external validation units would overstate the effective replication.

The fact that all 54 external test compounds were inside the HuSkin chemical domain is both a strength and a limitation. It strengthens the interpretation that the experiment tests **source transfer** rather than confounding source transfer with severe chemical extrapolation. Conversely, it means the present study provides no evidence that source calibration rescues predictions for compounds outside the molecular domain of the reference model. Chemical applicability and experimental-source applicability should therefore remain separate dimensions of deployment confidence.

5.5 Relation to previous work

Previous approaches have addressed permeability heterogeneity through standardization, latent common permeability, explicit experimental covariates, filtering, aggregation, and more recent record-level or mixed-effects analyses (Fujiwara et al. 2003; Abraham and Martins 2004; Samaras et al. 2012; Oh and Ahn 2026). The present framework is complementary rather than contradictory. Instead of attempting to remove all source variation before modeling, it retains measurement-level structure and asks whether the residual source component can be used operationally.

This distinction also separates the current analysis from studies whose primary objective is to maximize predictive performance from merged databases (Asgarkhanova et al. 2026; Oh and Ahn 2026). Pooling HuSkinDB and SkinPiX at the outset could produce a larger training set, but it would eliminate the clean cross-dataset transfer experiment. By preserving SkinPiX as an external source collection, the analysis tests a deployment behavior that would otherwise be difficult to identify.

5.6 Limitations

Several limitations constrain interpretation. First, the random reference effect is **source-associated**, not a causal estimate of laboratory, apparatus, operator, skin donor, or protocol effects individually. A publication-level intercept can absorb all systematic differences correlated with that source. The present results therefore do not justify attributing a fixed percentage of permeability variation causally to apparatus differences.

Second, a common random intercept assumes that a source shifts compounds approximately in parallel. The HUR_48 failure shows that this assumption can be violated. Future models could examine random slopes, protocol interactions, or hierarchical decomposition of source into laboratory, apparatus, tissue preparation,

and other metadata where sufficiently complete data exist. Such extensions should be justified by new data rather than added solely to improve retrospective fit.

Third, the empirical-Bayes calibration also uses an equal-anchor approximation after repeated measurements have been averaged within compound. Anchor compounds with different numbers of repeated measurements are therefore not weighted according to their measurement precision, and uncertainty in the training-derived compound effects is not propagated into the shrinkage calculation. A fully uncertainty-weighted formulation could incorporate both replicate-specific measurement precision and uncertainty in the estimated compound effects, but this would require a more elaborate calibration model than the simple deployment-oriented procedure evaluated here.

Fourth, the external SkinPiX validation is limited by source overlap. Although 54 novel compounds were evaluated, they came from only 4 eligible SkinPiX sources. The number of experimental sources, rather than the number of compound measurements alone, is therefore the main limitation on the generality of the cross-dataset conclusion.

Fifth, the chemical component was intentionally simple. The objective was to isolate the effect of measurement hierarchy and source calibration while holding molecular information relatively fixed, not to establish the highest-performing skin-permeability predictor available. More expressive molecular models could be substituted for $f(X)$, but they would require separate validation and should not be assumed to preserve the same calibration behavior.

Finally, the analyses are retrospective and rely on literature sources. The strongest validation would be prospective: fit and freeze the reference model, measure several chemically diverse anchor compounds on a previously unseen experimental platform, estimate its platform offset, and then predict additional compounds before observing their permeability. Such a study would directly test whether literature-derived source calibration transfers to a real new apparatus rather than another historical database.

5.7 Implications for experimental deployment

The framework suggests a practical deployment strategy. A reference model provides a molecular prediction on a common permeability scale. A new laboratory or platform then measures a small panel of anchor compounds already represented in the reference corpus. Their deviations from the reference expectation estimate a platform-specific offset, after shrinkage for limited anchor information. Predictions for new compounds can then be reported together with two separate applicability statements: whether the compound is chemically inside the reference domain and whether the experimental platform has been calibrated.

This creates four qualitatively different deployment states: (i) chemically in-domain and platform-calibrated, (ii) chemically in-domain but platform-uncalibrated, (iii) chemically out-of-domain but platform-calibrated, and (iv) chemically out-of-domain and platform-uncalibrated. The present study provides the strongest evidence for the first state and does not imply that source calibration compensates for chemical extrapolation.

6 Conclusions

Human skin permeability measurements contain substantial and reproducible source-associated variation that is not captured by molecular descriptors alone. Retaining measurement-level hierarchy does not consistently produce a better source-independent molecular QSPR, but it allows source context to be represented explicitly. When source-specific information is available from the reference data, incorporating that context materially improves prediction; when a source is new, shared anchor compounds can instead be used to estimate a source offset.

The calibration effect survived two important tests. It improved prediction of compounds absent from model fitting when entire HuSkin sources were held out, and in a pairwise double-holdout falsification, source-matched calibration consistently outperformed an otherwise equivalent offset estimated from another held-out source. The same calibration framework then transferred from HuSkinDB to SkinPiX measurements,

reducing mean source-level error for HuSkin-unseen compounds. These findings support a study-aware QSPR framework in which molecular prediction and experimental-source calibration are treated as distinct but complementary components of permeability prediction.

7 Data and code availability

The analysis uses publicly available datasets from Cheruvu et al. 2022, HuSkinDB, and SkinPiX (Cheruvu et al. 2022; Stepanov et al. 2020; Chedik et al. 2024). Analysis scripts and processed outputs are maintained in the project repository. The final repository release should include the exact raw-data provenance, software-environment information for RDKit descriptor generation, and all scripts required to regenerate manuscript tables and figures.

8 Author contributions

Chayada Piantham: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing.

Additional author contributions should be added according to the final author list and CRediT roles.

9 Funding

Funding information to be added before submission.

10 Competing interests

The author(s) declare no competing interests unless otherwise stated at submission.

11 Supplementary material

Table S1: Cheruvu hierarchical model comparison.

Model	AIC	BIC	Log-likelihood	df
M0	1033.64	1046.13	-513.82	3
M1	823.29	839.95	-407.64	4
M2	812.32	833.15	-401.16	5
M3	813.66	838.65	-400.83	6

Table S2: Prespecified molecular QSPR specifications and reduction in residual compound variance.

Specification	Compounds	Measurements	Baseline compound variance	Joint-model compound variance	Compound variance reduction (%)	Baseline AIC	Joint-model AIC
S1	148	476	0.691	0.274	60.3	812.32	704.52
S2	148	476	0.691	0.266	61.4	812.32	703.97

Table S3: Coefficient estimates for the conventional and hierarchical molecular QSPR specifications.

Specification	model	Term	Estimate	se
S1	Conventional	(Intercept)	-2.442	0.0628978
S1	Conventional	mw_z	-0.525	0.0643767
S1	Conventional	log_kow_z	0.636	0.0643767
S1	Hierarchical	(Intercept)	-2.612	0.0830520
S1	Hierarchical	temperature_c	0.038	0.0112105
S1	Hierarchical	mw_z	-0.370	0.0584279
S1	Hierarchical	log_kow_z	0.637	0.0536842
S2	Conventional	(Intercept)	-2.442	0.0594157
S2	Conventional	mw_z	-0.450	0.1426366
S2	Conventional	log_kow_z	0.326	0.1201162
S2	Conventional	melting_point_k_z	-0.238	0.0754869
S2	Conventional	log_saq_z	-0.213	0.1263555
S2	Conventional	h_donor_z	-0.245	0.0943533
S2	Conventional	h_acceptor_z	0.073	0.1317609
S2	Hierarchical	(Intercept)	-2.604	0.0819660
S2	Hierarchical	temperature_c	0.038	0.0111729
S2	Hierarchical	mw_z	-0.145	0.1266952
S2	Hierarchical	log_kow_z	0.500	0.1008058
S2	Hierarchical	melting_point_k_z	-0.156	0.0661164
S2	Hierarchical	log_saq_z	0.001	0.1059463
S2	Hierarchical	h_donor_z	-0.040	0.0841324
S2	Hierarchical	h_acceptor_z	-0.146	0.1117261

Table S4: Cheruvu leave-one-compound-out prediction performance.

Specification	Approach	RMSE	MAE	R2	r
S1	Conventional	0.866	0.690	0.389	0.630
S1	Hierarchical	0.721	0.557	0.576	0.761
S2	Conventional	0.850	0.674	0.411	0.649
S2	Hierarchical	0.691	0.534	0.610	0.784

Table S5: Cheruvu prediction error inside and outside the fold-specific chemical applicability domain.

Specification	In domain	Compounds	MAE conventional	MAE hierarchical
S1	Yes	137	0.622	0.506
S1	No	11	1.065	0.948
S2	Yes	128	0.578	0.484
S2	No	20	0.983	0.881

Table S6: Sensitivity analyses for Cheruvu leave-one-compound-out prediction.

Scenario	Specification	Approach	RMSE	MAE	R2	r
Primary S2	S2	Conventional	0.850	0.674	0.411	0.649
Primary S2	S2	Hierarchical	0.691	0.534	0.610	0.784
Multi-reference compounds	Multi-reference compounds	Conventional	0.952	0.737	0.287	0.674
Multi-reference compounds	Multi-reference compounds	Hierarchical	0.761	0.618	0.544	0.770
Exclude |standardized residual| > 3	Exclude |standardized residual| > 3	Conventional	0.840	0.668	0.421	0.657
Exclude |standardized residual| > 3	Exclude |standardized residual| > 3	Hierarchical	0.675	0.523	0.626	0.792
Mean descriptor aggregation	Mean descriptor aggregation	Conventional	0.850	0.675	0.410	0.649
Mean descriptor aggregation	Mean descriptor aggregation	Hierarchical	0.691	0.534	0.610	0.784
Expanded nonlinear specification	Expanded nonlinear specification	Conventional	0.733	0.560	0.562	0.750
Expanded nonlinear specification	Expanded nonlinear specification	Hierarchical	0.671	0.519	0.632	0.801

Table S7: Summary of hierarchical replication and secondary temperature analysis in HuSkinDB.

Metric	Value
Measurements	546
Compounds	251
Literature sources	95
AIC, compound-only hierarchy	1819.39
AIC, compound + source hierarchy	1622.44
Compound variance (%)	23.5
Source variance (%)	44.4
Residual variance (%)	32.1
Compound variance reduction by S1 (%)	37.1
Delta AIC after adding S1	-40.34
Temperature complete-case measurements	391
Temperature coefficient [log10(cm/h) per C]	-0.054
Temperature p-value	0.0383
Singular fit	No

Table S8: Held-out-source HuSkin calibration across all sources eligible at each anchor-panel size. The eligible source cohort changes as the required number of anchors increases.

Task	Anchors	Sources	RMSE uncal.	RMSE cal.	RMSE improvement (%)	MAE uncal.	MAE cal.	MAE improvement (%)
Known compound	1	43	0.973	0.887	8.9	0.880	0.801	8.9
Known compound	2	32	0.920	0.777	15.5	0.831	0.706	15.0
Known compound	3	22	1.008	0.780	22.6	0.905	0.698	22.9
Known compound	4	17	1.067	0.737	30.9	0.962	0.651	32.3
Known compound	5	14	1.119	0.727	35.0	1.032	0.656	36.5
New compound	1	31	1.235	1.125	8.9	1.156	1.038	10.1
New compound	2	22	1.166	0.982	15.8	1.095	0.898	17.9
New compound	3	19	1.001	0.796	20.6	0.921	0.721	21.7
New compound	4	12	1.096	0.701	36.0	1.006	0.636	36.8
New compound	5	9	1.135	0.621	45.3	1.049	0.568	45.8

Table S9: Source-level calibration results for the nine HuSkin sources supporting all anchor-panel sizes from one to five.

Source	Anchors	RMSE uncal.	RMSE cal.	RMSE improvement (%)	MAE improvement (%)
HUR_10	1	0.717	0.697	2.8	-2.1
HUR_11	1	1.444	0.987	31.7	33.4
HUR_14	1	0.510	0.440	13.6	2.1
HUR_34	1	1.062	0.844	20.5	20.5
HUR_39	1	1.050	0.652	37.9	41.5
HUR_48	1	0.600	0.720	-20.0	-48.7
HUR_5	1	2.641	1.276	51.7	58.7
HUR_62	1	1.347	0.740	45.1	47.9
HUR_82	1	0.848	0.682	19.6	15.8
HUR_10	2	0.717	0.677	5.5	0.4
HUR_11	2	1.444	0.863	40.2	42.6
HUR_14	2	0.510	0.402	21.0	11.7
HUR_34	2	1.062	0.818	23.0	23.0
HUR_39	2	1.050	0.631	39.9	42.9
HUR_48	2	0.600	0.791	-31.8	-65.4
HUR_5	2	2.641	1.003	62.0	65.3
HUR_62	2	1.347	0.573	57.5	61.1
HUR_82	2	0.848	0.590	30.4	26.4
HUR_10	3	0.717	0.647	9.7	4.5
HUR_11	3	1.444	0.805	44.3	47.0
HUR_14	3	0.510	0.374	26.7	14.3
HUR_34	3	1.062	0.829	21.9	21.9
HUR_39	3	1.050	0.633	39.7	43.0
HUR_48	3	0.600	0.828	-37.9	-72.5
HUR_5	3	2.641	0.901	65.9	69.7
HUR_62	3	1.347	0.510	62.2	65.6
HUR_82	3	0.848	0.551	35.0	28.2
HUR_10	4	0.717	0.647	9.8	4.4
HUR_11	4	1.444	0.754	47.8	50.8
HUR_14	4	0.510	0.367	28.1	15.2
HUR_34	4	1.062	0.763	28.2	28.2
HUR_39	4	1.050	0.542	48.4	52.5
HUR_48	4	0.600	0.872	-45.3	-80.7
HUR_5	4	2.641	0.831	68.5	72.9
HUR_62	4	1.347	0.467	65.4	68.8
HUR_82	4	0.848	0.526	37.9	30.9
HUR_10	5	0.717	0.639	10.9	5.1
HUR_11	5	1.444	0.732	49.3	52.5
HUR_14	5	0.510	0.350	31.3	15.4
HUR_34	5	1.062	0.764	28.1	28.1
HUR_39	5	1.050	0.513	51.2	55.4
HUR_48	5	0.600	0.882	-46.9	-82.1
HUR_5	5	2.641	0.758	71.3	72.9
HUR_62	5	1.347	0.445	67.0	69.9
HUR_82	5	0.848	0.509	40.0	31.7

Table S10: Chemical applicability-domain summary across Cheruvu, HuSkinDB, and the external SkinPiX transfer test. SkinPiX errors are not stratified because all external test compounds were in-domain.

Dataset	Specification	In-domain n	Out-of-domain n	MAE in domain	MAE out of domain
Cheruvu	S1	137	11	0.506	0.948
Cheruvu	S2	128	20	0.484	0.881
HuSkinDB	S1	243	8	0.807	2.137
SkinPiX external test	S1	54	0	–	–

Table S11: SkinPiX transfer and provenance audit summary.

Metric	Value
Retained measurements	199
Compounds	108
References	36
HuSkin-overlapping compounds	24
SkinPiX-only compounds	84
Eligible transfer references	4
References with DOI overlap	0
Median absolute RDKit MW difference among matched compounds	0
Median absolute RDKit logP difference among matched compounds	0

Table S12: Compound matches used to define HuSkinDB-SkinPiX overlap. Full SMILES and structure-level audit fields are retained in the machine-readable overlap file.

HuSkin ID	HuSkin compound	SkinPiX ID	SkinPiX compound	Match method
HUC_42	2-Nitro-p-Phenylenediamine	SPX_C5	2-Nitro-p-phenylenediamine	Name
HUC_40	Benzene	SPX_C19	Benzene	Name
HUC_49	Benzoic acid	SPX_C20	Benzoic acid	Name
HUC_2	Caffeine	SPX_C32	Caffeine	Name
HUC_153	Codeine	SPX_C39	Codeine	Name
HUC_246	Diethyl Phthalate	SPX_C43	Diethyl phthalate	Name
HUC_28	Diethylene glycol monobutyl ether	SPX_C44	Diethylene glycol monobutyl ether	SMILES
HUC_245	Dimethyl Phthalate	SPX_C47	Dimethyl phthalate	Name
HUC_159	Ethyl p-aminobenzoate	SPX_C49	Ethyl p-aminobenzoate	Name
HUC_127	Fentanyl	SPX_C52	Fentanyl	Name
HUC_103	Hydrocortisone	SPX_C56	Hydrocortisone	Name
HUC_154	Hydromorphone	SPX_C57	Hydromorphone	Name
HUC_80	Ibuprofen	SPX_C58	Ibuprofen	Name
HUC_85	Ketoprofen	SPX_C62	Ketoprofen	Name
HUC_160	Methyl p-aminobenzoate	SPX_C65	Methyl p-aminobenzoate	Name
HUC_156	Morphine	SPX_C68	Morphine	Name
HUC_82	Nicotine	SPX_C71	Nicotine	Name
HUC_111	Progesterone	SPX_C80	Progesterone	Name
HUC_58	Propylparaben	SPX_C83	Propylparaben	Name
HUC_96	Resorcinol	SPX_C84	Resorcinol	Name
HUC_60	Testosterone	SPX_C87	Testosterone	Name
HUC_89	Theophylline	SPX_C90	Theophylline	Name
HUC_235	Triclosan	SPX_C93	Triclosan	Name
HUC_46	p-Phenylenediamine	SPX_C102	p-Phenylenediamine	Name

Table S13: Cross-dataset SkinPiX calibration results by source and anchor-panel size.

Source	Anchors	RMSE uncal.	RMSE cal.	RMSE improvement (%)	MAE improvement (%)
SPX_R1	1	1.159	0.715	38.3	44.9
SPX_R18	1	0.663	0.708	-6.8	-2.2
SPX_R4	1	0.393	0.352	10.5	21.7
SPX_R5	1	1.025	0.895	12.7	13.6
SPX_R1	2	1.159	0.657	43.3	50.4
SPX_R18	2	0.663	0.677	-2.1	-3.9
SPX_R5	2	1.025	0.885	13.7	14.7
SPX_R1	3	1.159	0.623	46.2	53.5
SPX_R18	3	0.663	0.658	0.8	-3.2
SPX_R5	3	1.025	0.885	13.7	16.6
SPX_R1	4	1.159	0.615	46.9	54.3
SPX_R5	4	1.025	0.878	14.3	18.3
SPX_R1	5	1.159	0.599	48.3	55.7

Table S14: Target-source results from the pairwise double-holdout falsification analysis. Results are averaged across eligible donor sources within each target source before cohort-level aggregation.

Target source	Anchors	Donor sources	Test n	RMSE uncal.	RMSE matched	RMSE different	Matched improvement (%)	Different improvement (%)	Different - matched RMSE
HUR_1	1	62	1	1.861	1.606	1.852	13.7	0.5	0.246
HUR_10	1	62	7	0.717	0.696	0.868	3.0	-21.0	0.172
HUR_11	1	62	6	1.445	0.986	1.453	31.8	-0.5	0.467
HUR_13	1	61	8	1.301	1.339	1.418	-2.9	-9.0	0.079
HUR_14	1	61	2	0.510	0.441	0.674	13.6	-32.0	0.233
HUR_15	1	61	5	0.899	0.846	1.047	5.8	-16.5	0.201
HUR_17	1	62	4	0.619	0.683	0.810	-10.3	-30.9	0.127
HUR_18	1	62	2	1.931	2.180	2.014	-12.9	-4.3	-0.165
HUR_2	1	62	14	1.382	1.870	1.423	-35.3	-2.9	-0.447
HUR_20	1	62	5	0.695	0.854	0.877	-22.9	-26.2	0.023
HUR_22	1	62	2	4.308	4.116	4.310	4.5	0.0	0.194
HUR_24	1	62	1	0.604	0.721	0.713	-19.3	-18.0	-0.008
HUR_27	1	61	1	0.026	0.747	0.445	-2803.9	-1630.5	-0.302
HUR_3	1	61	5	1.217	1.076	1.312	11.5	-7.8	0.236
HUR_31	1	62	13	1.083	1.223	1.216	-13.0	-12.3	-0.007
HUR_34	1	62	1	1.063	0.863	1.081	18.8	-1.8	0.218
HUR_37	1	62	1	0.102	0.468	0.453	-358.1	-343.8	-0.015
HUR_39	1	61	10	1.051	0.712	1.111	32.2	-5.8	0.399
HUR_40	1	62	1	0.459	0.072	0.587	84.2	-28.0	0.515
HUR_41	1	61	3	1.153	1.314	1.266	-13.9	-9.7	-0.048
HUR_48	1	62	7	0.601	0.711	0.795	-18.4	-32.3	0.083
HUR_5	1	61	2	2.642	1.350	2.630	48.9	0.5	1.280
HUR_51	1	61	3	1.245	0.980	1.273	21.3	-2.2	0.293
HUR_52	1	61	9	0.605	0.668	0.797	-10.4	-31.9	0.130
HUR_57	1	61	8	0.917	0.574	0.979	37.4	-6.7	0.404
HUR_62	1	62	4	1.348	0.750	1.367	44.4	-1.4	0.618
HUR_78	1	62	2	2.833	2.470	2.846	12.8	-0.5	0.376
HUR_82	1	62	3	0.848	0.663	0.958	21.8	-13.0	0.295
HUR_83	1	62	1	2.181	1.797	2.191	17.6	-0.5	0.393
HUR_90	1	62	3	2.117	1.504	2.130	28.9	-0.6	0.626
HUR_94	1	62	4	0.518	0.694	0.659	-34.0	-27.2	-0.035
HUR_1	2	42	1	1.863	1.535	1.799	17.6	3.4	0.264

HUR_10	2	42	7	0.718	0.674	0.886	6.1	-23.4	0.211
HUR_11	2	42	6	1.447	0.855	1.427	40.9	1.4	0.572
HUR_14	2	41	2	0.511	0.405	0.694	20.8	-35.7	0.289
HUR_17	2	40	4	0.617	0.685	0.843	-11.0	-36.6	0.158
HUR_18	2	41	2	1.931	2.263	2.052	-17.2	-6.3	-0.211
HUR_20	2	42	5	0.694	0.907	0.928	-30.7	-33.7	0.021
HUR_24	2	42	1	0.604	0.752	0.794	-24.5	-31.4	0.042
HUR_31	2	41	13	1.082	1.135	1.270	-4.9	-17.3	0.134
HUR_34	2	42	1	1.065	0.806	1.064	24.3	0.1	0.258
HUR_37	2	42	1	0.103	0.479	0.503	-363.9	-387.9	0.025
HUR_39	2	41	10	1.053	0.621	1.084	41.1	-2.9	0.463
HUR_41	2	41	3	1.153	1.238	1.329	-7.4	-15.3	0.091
HUR_48	2	42	7	0.600	0.786	0.848	-30.9	-41.2	0.062
HUR_5	2	42	2	2.644	1.043	2.577	60.5	2.5	1.534
HUR_51	2	42	3	1.244	0.887	1.390	28.7	-11.8	0.503
HUR_52	2	41	9	0.605	0.628	0.836	-3.9	-38.2	0.208
HUR_62	2	42	4	1.350	0.587	1.340	56.5	0.7	0.753
HUR_78	2	42	2	2.832	2.370	2.925	16.3	-3.3	0.555
HUR_82	2	41	3	0.846	0.594	1.033	29.7	-22.1	0.439
HUR_83	2	42	1	2.178	1.684	2.270	22.7	-4.2	0.586
HUR_94	2	42	4	0.520	0.725	0.663	-39.6	-27.5	-0.063
HUR_10	3	30	7	0.719	0.662	0.876	7.9	-22.0	0.215
HUR_11	3	31	6	1.448	0.797	1.397	45.0	3.5	0.600
HUR_14	3	30	2	0.513	0.381	0.684	25.7	-33.4	0.303
HUR_17	3	30	4	0.617	0.686	0.867	-11.3	-40.5	0.180
HUR_20	3	31	5	0.694	0.935	0.927	-34.7	-33.5	-0.009
HUR_24	3	31	1	0.605	0.769	0.806	-27.2	-33.3	0.037
HUR_31	3	31	13	1.082	1.082	1.287	0.0	-18.9	0.204
HUR_34	3	31	1	1.067	0.789	1.035	26.1	3.0	0.246
HUR_37	3	30	1	0.098	0.519	0.426	-428.9	-334.0	-0.093
HUR_39	3	30	10	1.055	0.584	1.066	44.7	-1.0	0.481
HUR_41	3	31	3	1.153	1.209	1.356	-4.9	-17.6	0.147
HUR_48	3	31	7	0.600	0.835	0.854	-39.1	-42.2	0.019
HUR_5	3	30	2	2.646	0.914	2.511	65.5	5.1	1.597
HUR_51	3	31	3	1.245	0.854	1.439	31.4	-15.6	0.585
HUR_52	3	30	9	0.605	0.593	0.827	2.0	-36.6	0.234
HUR_62	3	31	4	1.352	0.518	1.320	61.7	2.4	0.802
HUR_82	3	30	3	0.844	0.556	1.051	34.1	-24.6	0.495
HUR_83	3	31	1	2.177	1.635	2.369	24.9	-8.8	0.734

HUR_94	3	30	4	0.521	0.744	0.649	-42.8	-24.6	-0.095
HUR_10	4	21	7	0.719	0.650	0.953	9.6	-32.6	0.303
HUR_11	4	21	6	1.450	0.760	1.428	47.6	1.5	0.668
HUR_14	4	20	2	0.514	0.363	0.766	29.4	-48.9	0.403
HUR_24	4	21	1	0.606	0.780	0.869	-28.7	-43.4	0.089
HUR_31	4	20	13	1.083	1.043	1.359	3.7	-25.5	0.316
HUR_34	4	21	1	1.070	0.770	1.082	28.0	-1.2	0.312
HUR_39	4	20	10	1.057	0.555	1.151	47.5	-8.9	0.596
HUR_48	4	21	7	0.600	0.857	0.921	-42.8	-53.4	0.064
HUR_5	4	20	2	2.648	0.826	2.504	68.8	5.5	1.678
HUR_51	4	21	3	1.246	0.824	1.508	33.9	-21.0	0.683
HUR_62	4	20	4	1.354	0.472	1.378	65.2	-1.8	0.906
HUR_82	4	20	3	0.844	0.528	1.051	37.4	-24.6	0.523
HUR_10	5	16	7	0.721	0.644	1.005	10.6	-39.5	0.361
HUR_11	5	16	6	1.452	0.739	1.400	49.1	3.6	0.660
HUR_14	5	16	2	0.516	0.351	0.836	32.0	-62.1	0.485
HUR_34	5	16	1	1.072	0.767	1.064	28.4	0.7	0.297
HUR_39	5	15	10	1.059	0.533	1.177	49.7	-11.1	0.644
HUR_48	5	16	7	0.600	0.878	1.022	-46.4	-70.3	0.144
HUR_5	5	15	2	2.650	0.767	2.413	71.1	9.0	1.646
HUR_62	5	15	4	1.356	0.447	1.372	67.1	-1.1	0.925
HUR_82	5	15	3	0.842	0.510	1.149	39.5	-36.4	0.640

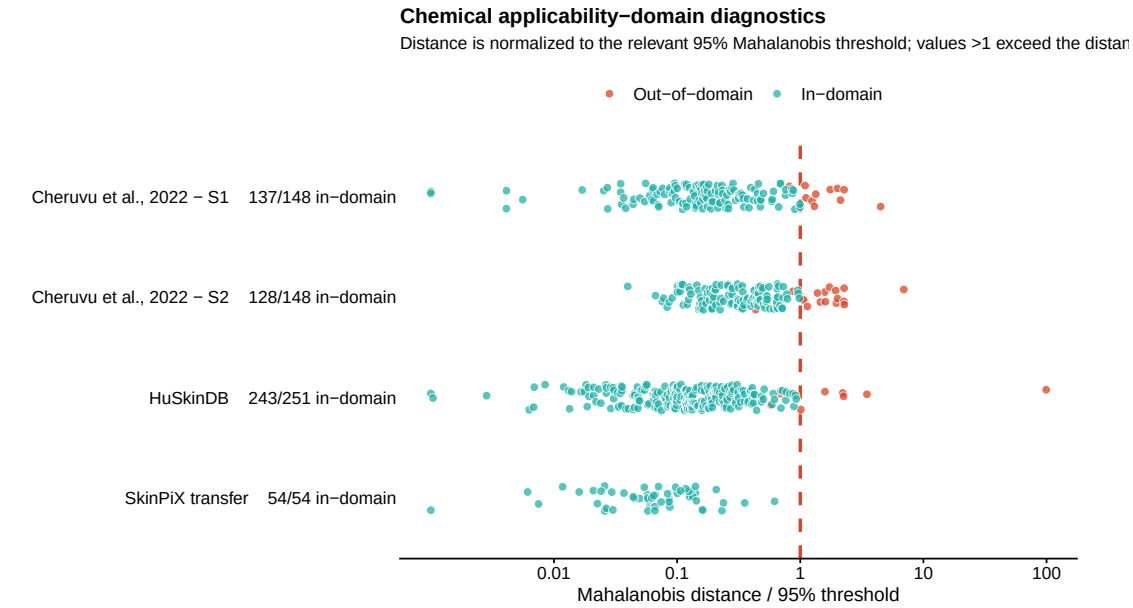


Figure S1: Applicability-domain diagnostics across Cheruvu S1, Cheruvu S2, HuSkinDB, and the external SkinPiX transfer set. Mahalanobis distance is normalized to the relevant 95% threshold; classification additionally requires all descriptors to remain within their training-set univariate ranges.

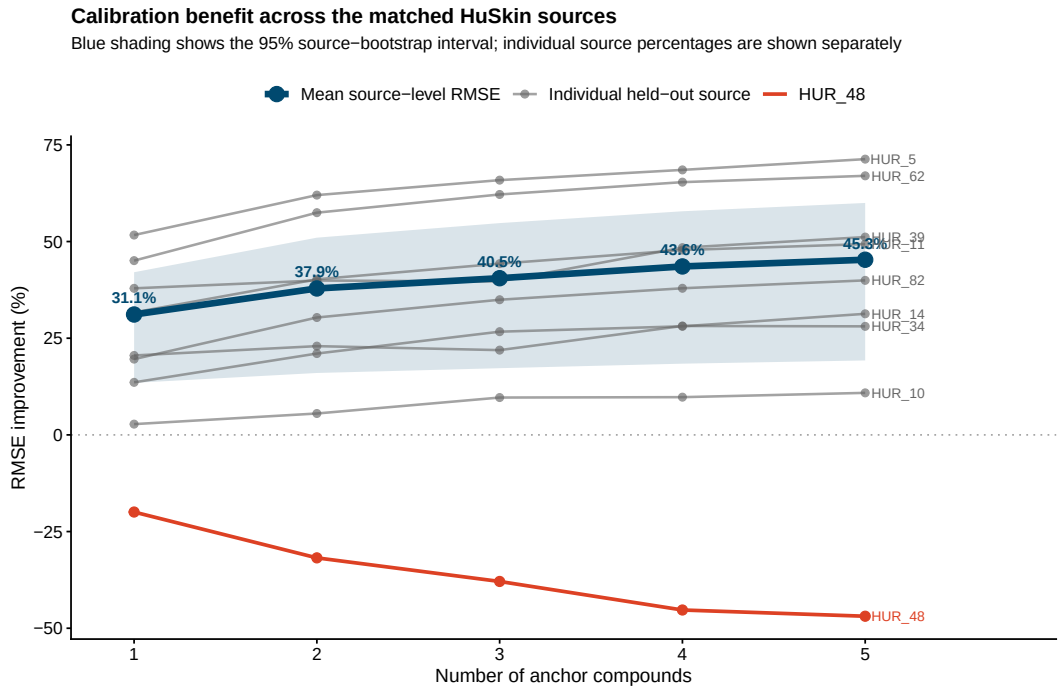


Figure S2: Calibration benefit across the nine matched HuSkin sources. Individual trajectories show source-specific RMSE improvement across anchor-panel sizes, while the cohort-level summary represents the ratio of mean calibrated to mean uncalibrated source-level RMSE rather than the arithmetic mean of source-specific percentage improvements. HUR_48 is retained as a persistent calibration failure.

References

Abraham, Michael H., and Filomena Martins. 2004. "Human Skin Permeation and Partition: General Linear Free-Energy Relationship Analyses." *Journal of Pharmaceutical Sciences* 93 (6): 1508–23. https://doi.org/10.1002/jps.20070.

Asgarkhanova, Farah, Shamkhal Baybekov, Gilles Marcou, et al. 2026. "Current Insights on Skin Permeability Data and Quantitative Structure-Property Relationship Modeling." *Molecular Informatics* 45 (4): e70030. https://doi.org/10.1002/minf.70030.

Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using Lme4." *Journal of Statistical Software* 67 (1): 1–48. https://doi.org/10.18637/jss.v067.i01.

Chedik, Lisa, Shamkhal Baybekov, Frédéric Cosnier, Gilles Marcou, Alexandre Varnek, and Catherine Champmartin. 2024. "An Update of Skin Permeability Data Based on a Systematic Review of Recent Research." *Scientific Data* 11: 224. https://doi.org/10.1038/s41597-024-03026-4.

Cheruvu, Hanumanth Srikanth, Xin Liu, Jeffrey E. Grice, and Michael S. Roberts. 2022. "An Updated Database of Human Maximum Skin Fluxes and Epidermal Permeability Coefficients for Drugs, Xenobiotics, and Other Solutes Applied as Aqueous Solutions." *Data in Brief* 42: 108242. https://doi.org/10.1016/j.dib.2022.108242.

Fujiwara, Shin-Ichi, Fumiyoshi Yamashita, and Mitsuru Hashida. 2003. "QSAR Analysis of Interstudy Variable Skin Permeability Based on the 'Latent Membrane Permeability' Concept." *Journal of Pharmaceutical Sciences* 92 (10): 1939–46. https://doi.org/10.1002/jps.10462.

Landrum, Greg et al. 2026. *RDKit: Open-Source Cheminformatics*. https://www.rdkit.org/.

Oh, Seunghwan, and So Hyun Ahn. 2026. "Human Skin Permeability Prediction by Machine Learning for Transdermal Drug Delivery." *Medicina* 62 (10): 1929. https://doi.org/10.3390/medicina62101929.

Potts, Russell O., and Richard H. Guy. 1992. "Predicting Skin Permeability." *Pharmaceutical Research* 9 (5): 663–69. https://doi.org/10.1023/A:1015810312465.

Samaras, Eleftherios G., Jim E. Riviere, and Taravat Ghafourian. 2012. "The Effect of Formulations and Experimental Conditions on in Vitro Human Skin Permeation—Data from Updated EDETOX Database." *International Journal of Pharmaceutics* 434 (1-2): 280–91. https://doi.org/10.1016/j.ijpharm.2012.05.012.

Stepanov, Dmitri, Steven Canipa, and Gerhard Wolber. 2020. "HuskinDB, a Database for Skin Permeation of Xenobiotics." *Scientific Data* 7: 426. https://doi.org/10.1038/s41597-020-00764-z.

Tsakovska, Ivanka, Ilza Pajeva, Merilin Al Sharif, et al. 2017. "Quantitative Structure–Skin Permeability Relationships." *Toxicology* 387: 27–42. https://doi.org/10.1016/j.tox.2017.06.008.

Waters, Laura J., and Xin Ling Quah. 2022. "Predicting Skin Permeability Using HuskinDB." *Scientific Data* 9: 584. https://doi.org/10.1038/s41597-022-01698-4.

Zeng, Ruolan, Jiyong Deng, Limin Dang, and Xinliang Yu. 2021. "Correlation Between the Structure and Skin Permeability of Compounds." *Scientific Reports* 11: 10076. https://doi.org/10.1038/s41598-021-89587-5.